FamilyForge tiering logic (acmg.tier_candidate) — evaluated top-to-bottom; first match wins

Candidate variant
(model + annotations attached)

Post-hoc caps:
• ClinVar conflicting + Tier A -> Tier B
• low genotype confidence / MZ-twin discordance + Tier A -> Tier B

Rare enough for its model?
(dominant: max AF ≤ af_dominant_max
recessive: max AF ≤ af_recessive_max)
AND ClinVar not benign

yes

ClinVar
pathogenic / likely-pathogenic?

no

Biallelic predicted-LoF in a
LoF-mechanism gene? (recessive model)

no

STAT3 dominant-negative
hotspot (DBD/SH2) missense/LoF
under de-novo/dominant?

no

Rare damaging missense in a
Tier-1/2 gene + strong in-silico
(AlphaMissense LP / REVEL≥0.7 / CADD≥25)
under a matching model?

no

Predicted-LoF in a panel gene?

no

Rare missense + supportive in-silico
(SIFT/PolyPhen/PhyloP/REVEL≥0.5)?

no

TIER C
rare in-panel, weaker support (or downgraded:
common in gnomAD / ClinVar benign)

yes

TIER B
strong VUS (mechanism / inheritance to confirm)

yes

yes

yes

yes

yes

yes

TIER A
likely actionable / likely-pathogenic candidate

no